

Authors

- Dubon Rodrigue^{2,3}
- Roberto Boghetti^{1,4}
- Jérôme H. Kämpf^{1,4}
- Mohamed T. Mabrouk²
- Bastien Padeloup³
- Patrick Meyer³
- Bruno Lacarrière²

Keywords

- District heating
- Physical modeling
- Dynamic simulation tools
- Open-access software

Affiliations

¹Idiap Research Institute


²GEPEA (CNRS 6144)


³Lab-STICC (CNRS 6285)

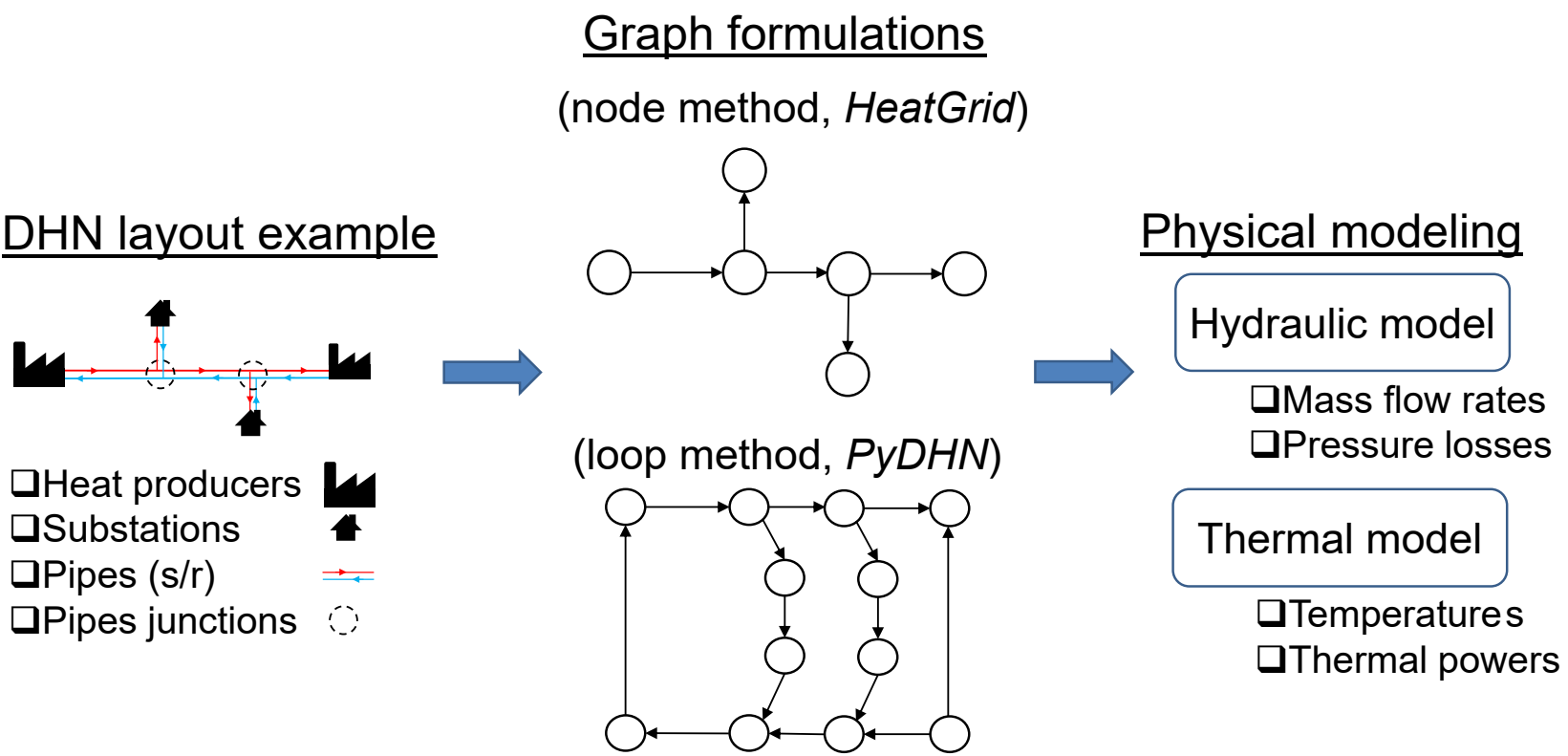

⁴EPFL, L'IDIAP


Let's discuss further



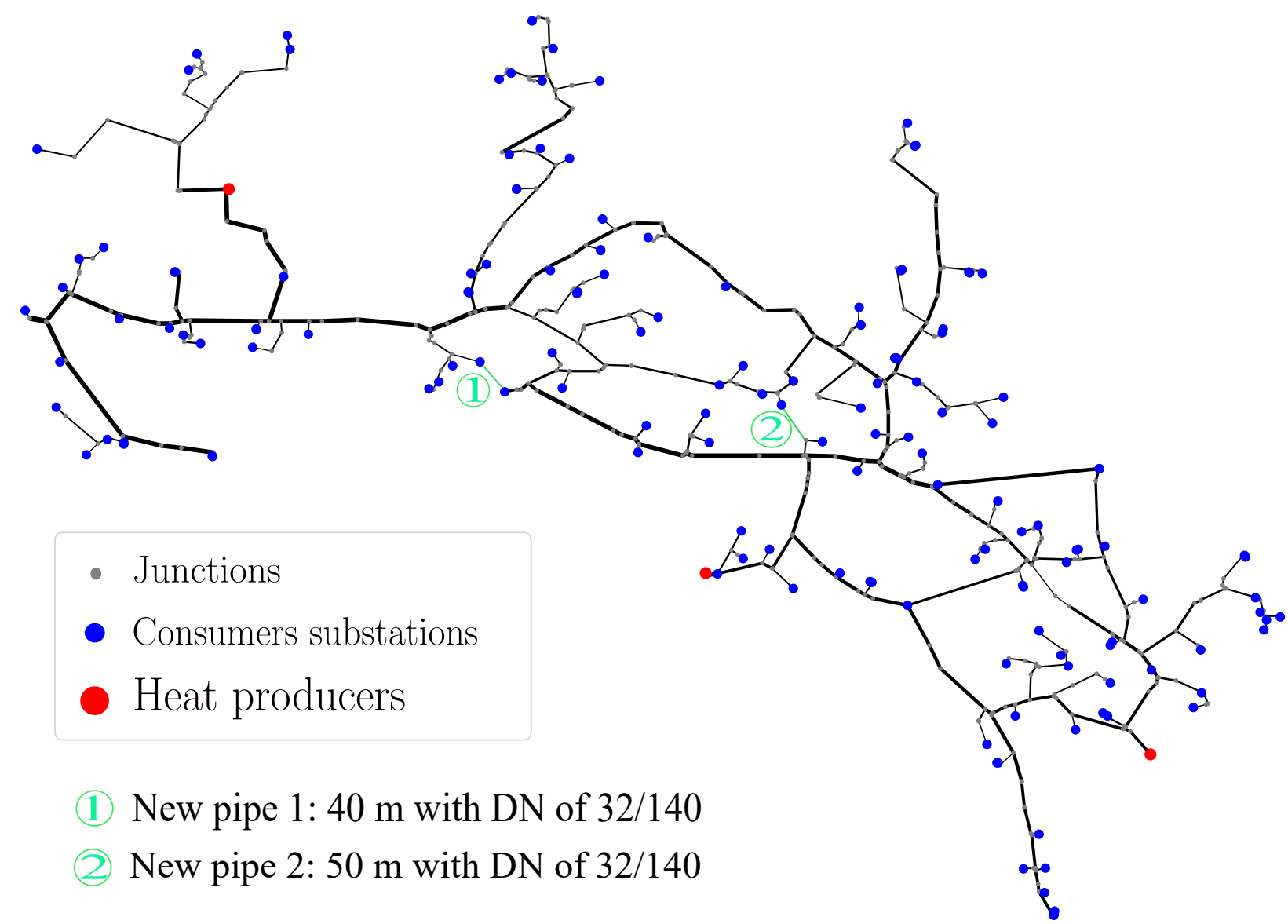
1 Context

- ❑ District Heating Networks (DHNs) may offer sustainable solution to distribute heat.
- ❑ An accurate digital twin of the DHN requires a validated and precise simulation model.



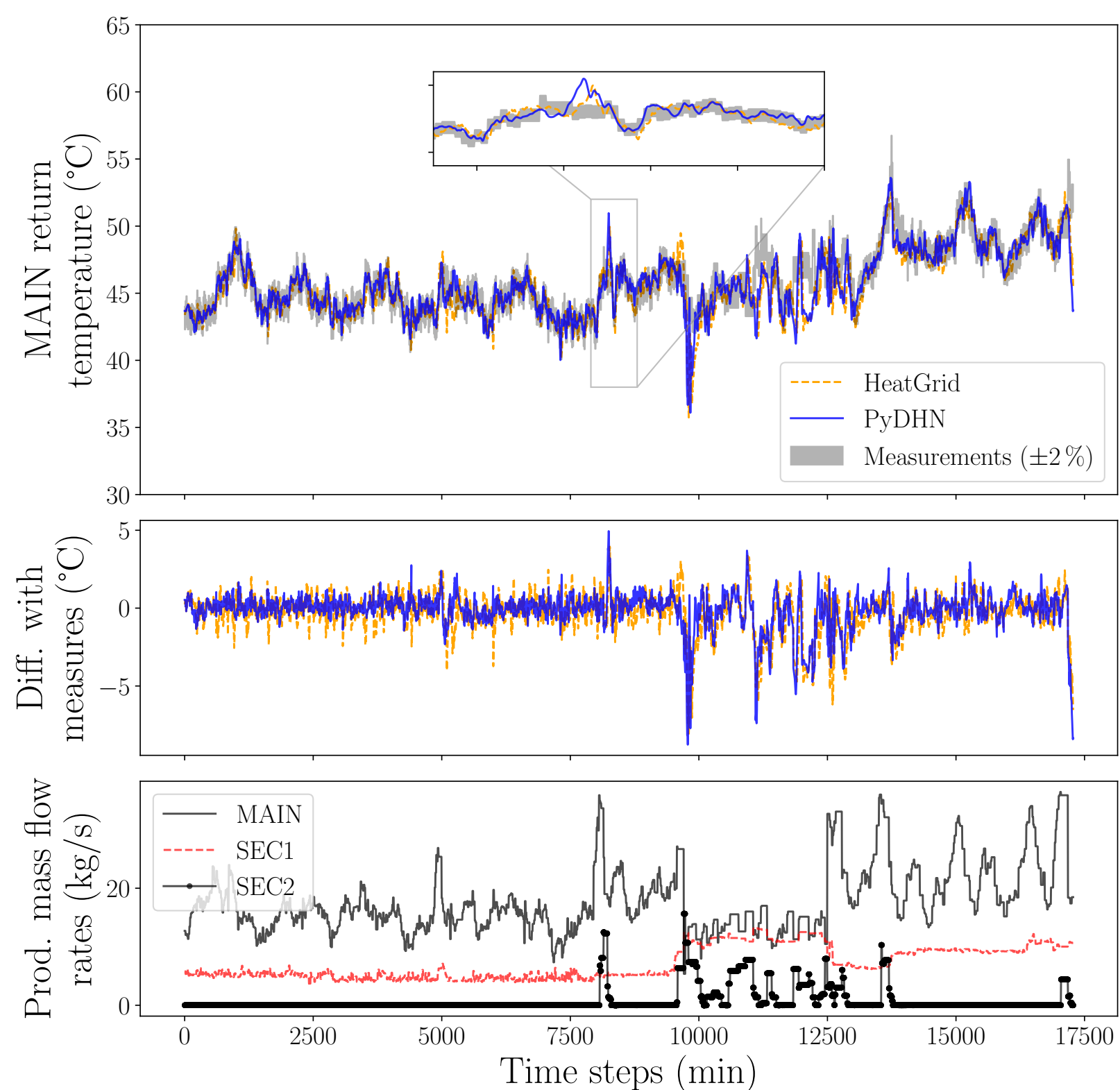
3 Case study DHN

Verbier DHN (**base**): 159 substations, 3 heat producers (MAIN, SEC1, SEC2) and 28 km of pipes



5 Validation results

5.1 Thermal validation



Return temperature diff.	Absolute	Relative
HeatGrid vs Measures	0.89 °C	1.93 %
PyDHN vs Measures	0.73 °C	1.57 %
HeatGrid vs PyDHN	0.75 °C	1.66 %

Papers: cv.hal.science/dubon-rodrigue

2 Objectives

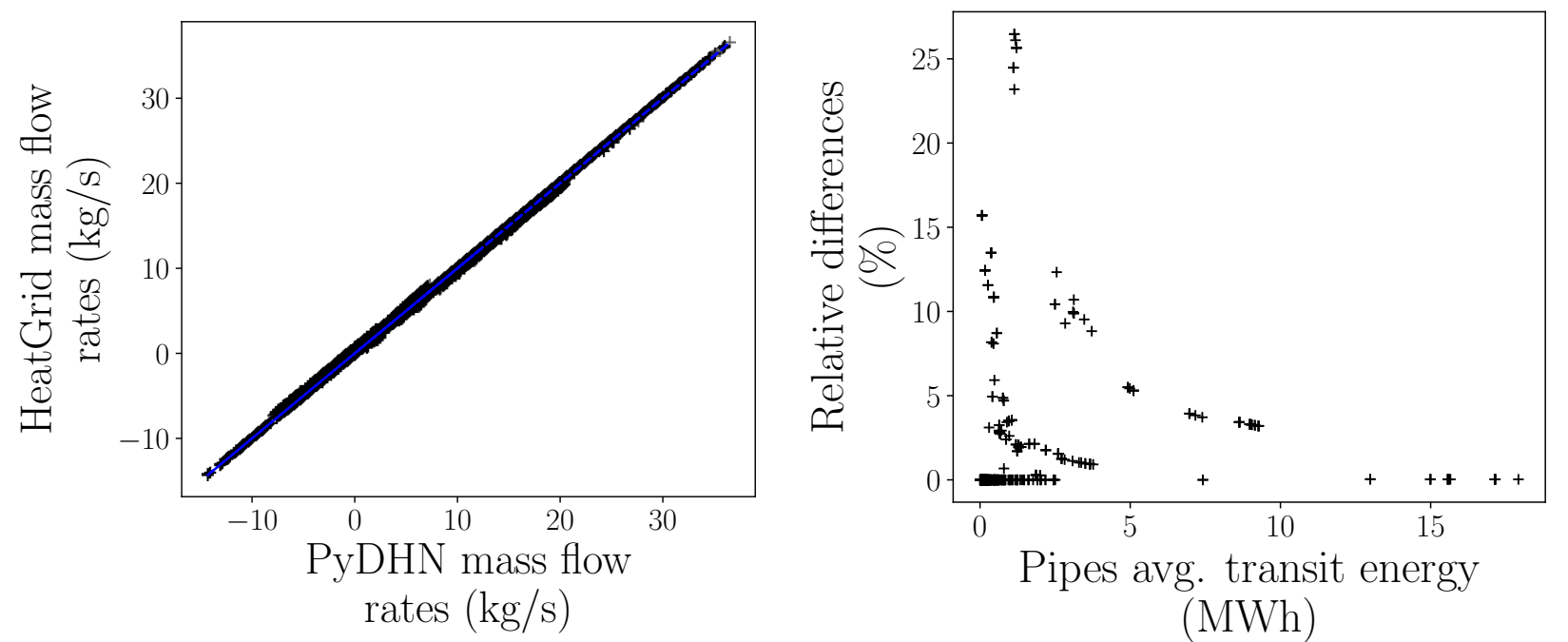
- ❑ **Validation of two distinct simulation models** (*i.e.*, *HeatGrid* and *PyDHN*) against measurements
- ❑ Evaluation of adding **new looping pipes** performances using these models

Model	Graph formulation		Physical modeling	
	Nodes	Edges	Hydraulic	Thermal
<i>HeatGrid</i>	Producers, Junctions, Substations	Pipes	Node method	Heat Transport Equation (Euler explicit)
<i>PyDHN</i>	Junctions	Producers, Junctions, Substations	Adapted loop method	Two-capacity model (Lagrangian approach)

4 Experimental setup and KPIs

- ❑ **Validation:**
 - Models validated on Verbier DHN (**base**)
 - Simulations results compared with measures
 - *Hydraulic KPI*: Avg. diff. in mass flow rates between models, averaged over network and time
 - *Thermal KPI*: Avg. diff. in return temperature at MAIN heat producer (models vs. measurements), over time.
- ❑ **Evaluation of new pipes:**
 - Scenario 1: Verbier DHN (**base**) + (1)
 - Scenario 2: Verbier DHN (**base**) + (2)
 - Scenario 3: Verbier DHN (**base**) + (1) + (2)
 - *KPIs*: changes in pumping costs and thermal losses (*i.e.*, + **increase** or - **decrease**)
 - *Costs*: electricity (25 cts CHF/kWh), heat (6 cts CHF/kWh), installation of pipe (1000 CHF/meter)

5.2 Hydraulic validation



Mass flow rates diff.	Absolute	Relative
HeatGrid vs PyDHN	0.0265 kg/s	1.53 %

6 Effects of new pipes results

Scenario	Model	Thermal loss (kWh)	Pumping energy (kWh)	Operating cost (CHF)
Scenario 1	<i>HeatGrid</i>	+206 (0.18%)	-12.7 (3.15%)	+15.5 (0.23%)
	<i>PyDHN</i>	+344 (0.33%)	-16.7 (2.97%)	+16.5 (0.27%)
Scenario 2	<i>HeatGrid</i>	+359 (0.32%)	-0.7 (0.20%)	+24.8 (0.37%)
	<i>PyDHN</i>	+404 (0.39%)	-0.9 (0.15%)	+25.0 (0.41%)
Scenario 3	<i>HeatGrid</i>	+499 (0.45%)	-13.2 (3.91%)	+30.1 (0.45%)
	<i>PyDHN</i>	+683 (0.67%)	-17.4 (3.10%)	+36.6 (0.60%)

7 Conclusion and Perspectives

- ✓ Both models effectively capture the dynamics of DHNs.
- ✓ Adding the pipes resulted in no savings in operating costs but add some flexibility in the network.
- ✓ Exploring **Machine Learning** to reduce simulation computational costs